

Annanahmed Shaikh

Boston, MA | (617) 407-4841 | annan.shaikh0404@gmail.com
linkedin.com/in/annanahmed-shaikh | github.com/annanshaikh04 | annanshaikh04.github.io

PROFESSIONAL SUMMARY

Machine Learning & Data Analytics Engineer with 1.5+ years of experience specialized in building scalable AI-driven data systems. Expert in fine-tuning Transformers for mathematical reasoning and optimizing high-concurrency database schemas (10K+ transactions/day). Proven track record of delivering 40% performance gains and achieving 0.86 ROC-AUC in large-scale multi-modal predictive modeling.

EDUCATION

Wentworth Institute of Technology
Master of Science in Data Science, GPA: 4.0/4.0
Silver Oak College of Engineering & Technology
Bachelor of Engineering in Computer Science

Boston, MA
Sept. 2024 – Apr. 2026
Ahmedabad, India
July 2019 – May 2023

PROFESSIONAL EXPERIENCE

Research Assistant - AI & LLM Systems
Wentworth Institute of Technology & Amazon Collaboration

Jan. 2026 – Mar. 2026
Boston, MA

- **Spearheaded** development of State-Space Models (S5, Mamba) using PyTorch for deciphering multi-terabyte biosignal datasets on Unity HPC clusters.
- **Orchestrated** advanced signal denoising and feature extraction pipelines, reducing model error rates by 15%; research accepted for publication at INTERSPEECH 2026.

Software Developer Intern
N & T Software Pvt. Ltd.

Nov. 2023 – Jul. 2024
Ahmedabad, India

- **Architected** 15+ RESTful APIs (Python, Node.js) contributing to a 40% reduction in latency for 10K+ daily transactions via connection pooling and Redis caching.
- **Leveraged** 3NF normalization and multi-key indexing to redesign legacy MongoDB/PostgreSQL schemas, achieving a 35% boost in complex query performance.
- **Automated** multi-source ETL workflows using pandas and APScheduler, eliminating 88% of manual processing overhead for operational analytics dashboards.

Full Stack Developer Intern
Infosense Services

Jan. 2023 – Jun. 2023
Gandhinagar, India

- **Pioneered** a real-time incident analytics system handling 5K+ daily events; deployed Celery task queues to ensure zero-loss log persistence during high-concurrency peak intervals.
- **Augmented** data retrieval speed by 60% through query plan optimization and horizontal microservice scaling using Docker and CI/CD pipelines.

PROJECTS (ACADEMIC & RESEARCH)

VentureFlow AI: Multi-Embedding Prediction Platform | Capstone 1

2025

- **Technical Approach:** Fused high-dimensional dual-LLM embeddings (MPNet, MiniLM) with structured venture capital data warehouses (46K+ records) using gradient-boosted models.
- **Quantifiable Result:** Achieved a validated **0.86 ROC-AUC**, successfully forecasting exits for 22K+ high-potential companies; awarded ****Winner**** for Industry Selection Showcase.

EV Charging Load Forecasting & Infrastructure | Applied Statistics

2024

- **Engineering:** Developed a timezone-safe data pipeline for 7,500+ charger sessions; implemented proportional overlap allocation for precise load profiling via ARIMA and Random Forest.
- **Impact:** Identified peak-demand windows suggesting **20% grid cost-savings**, enabling data-driven load-shifting strategies for energy regulators.

Multi-Step Reasoning Logic Enhancement | Capstone 2 (Dissertation)

2026

- **Innovation:** Designed a custom attention-reweighting layer for frozen transformers to rectify logic propagation errors in multi-step mathematical instructions.
- **Success:** Increased logic propagation accuracy from ****30% to 42% (12% absolute gain)**** without increasing computational latency vs. vanilla LLM baselines.

Transformer-based Credibility Classifier | Machine Learning

2025

- **Implementation:** Fine-tuned Sentence Transformers with UMAP dimensionality reduction to cluster latent linguistic patterns; integrated SHAP for explainable safety-critical news monitoring.

ACHIEVEMENTS

Published Researcher @ INTERSPEECH 2026

Awarded for novel signal decoding research at the world's premier venue for speech and signal processing; utilized Unity HPC and Mamba architectures.

TECHNICAL SKILLS

Machine Learning: PyTorch, Transformers, Hugging Face, LLM Fine-Tuning (LoRA), Scikit-learn, XGBoost, SHAP, Mamba, S5
Data Engineering: SQL (Expert), PostgreSQL, MongoDB, Redis, ETL Pipelines, 3NF/Star Schema, Parquet, Arrow, APScheduler
Architecture & Cloud: AWS (EC2, S3), RESTful APIs, Docker, CI/CD, Celery, Microservices, Unity HPC, Linux, Jupyter
Analytics & Tools: Tableau, Plotly, Streamlit, R, Java, JavaScript, C++, Git & GitHub, Excel (Adv), Hypothesis Testing